Yes. Based on the material you uploaded, the topic is **Probability**, especially its foundations, conditional probability, independence, Bayes' theorem, and their role in machine learning. The source explicitly frames probability as the language for representing uncertainty in ML rather than merely producing a class prediction.

I'll turn it into **interview-focused 80/20 notes** rather than reproducing the chapter. The goal is: **understand → visualize → implement → solve interview questions → connect to ML**.

Probability — 80/20 Interview Notes

1. What is Probability?

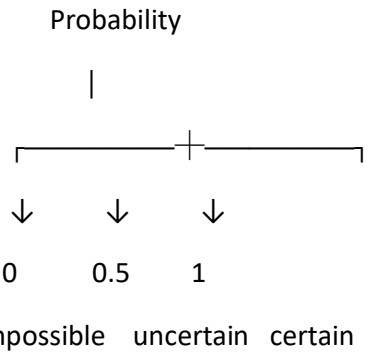
Definition

Probability is a numerical measure of uncertainty about an event, ranging from 0 to 1.

- 0 → impossible
- 1 → certain
- 0 < P(A) < 1 → uncertain

The source describes probability as a way to move beyond classical true/false reasoning and represent degrees of uncertainty.

Basic idea



ML connection

Instead of:

Model → Spam

we want:

Model → P(Spam | Email features) = 0.93

This tells us **how strongly the model believes** the email is spam.

The source explicitly connects ML prediction with estimating probabilities such as P(label | data).

---

2. Experiment, Outcome, Sample/Event Space, Event

These are fundamental vocabulary questions.

Experiment

An action/process whose result is uncertain.

Example:

Roll a die

Outcome

One possible result.

1, 2, 3, 4, 5, 6

Sample/Event Space — Ω

The set of all possible outcomes.

Ω = {1,2,3,4,5,6}

The source defines the event space as the collection of possible mutually exclusive outcomes.

Event

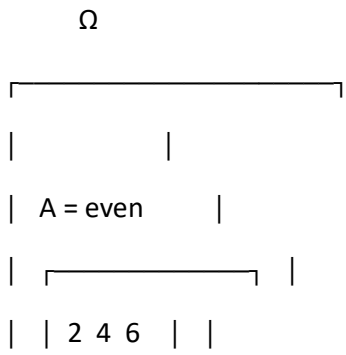
A subset of the sample space.

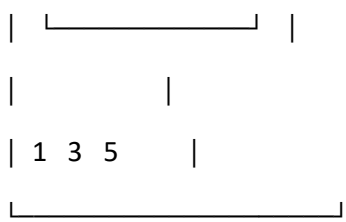
For example:

A = "even number"

A = {2,4,6}

Diagram





Interview question

Q: What is the difference between an outcome and an event?

Answer:

An outcome is one possible result of an experiment, while an event is a set of one or more outcomes.

---

3. Probability Axioms

This is one of the highest-value theoretical areas.

A valid probability measure satisfies:

Axiom 1

$$P(\Omega) = 1$$

Something from the sample space must occur.

Axiom 2

$$P(\emptyset) = 0$$

The impossible event has probability zero.

Axiom 3 — Additivity

For mutually exclusive events:

$$P(A \cup B) = P(A) + P(B)$$

More generally, for disjoint events:

$$P\left(\bigcup_i A_i\right) = \sum_i P\left(A_i\right)$$

The uploaded source defines probability measures using exactly these properties, including σ-additivity.

Example

Roll a die.

A = {1}

B = {2}

They cannot happen simultaneously.

Therefore:

$$P(A \cup B) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

---

4. Complement Rule

If A is an event:

$$P(A^c) = 1 - P(A)$$

Example

Probability of getting at least one 6 in one roll:

$$P(6) = \frac{1}{6}$$

Probability of **not** getting 6:

$$P(\text{not } 6) = 1 - \frac{1}{6} = \frac{5}{6}$$

Interview shortcut

Whenever you see:

"at least one"

think:

P(at least one)

=

1 - P(none)

Example

Probability of at least one head in 3 fair coin tosses:

$$\begin{aligned} & 1 - P(TTT) \\ & = 1 - \left(\frac{1}{2}\right)^3 \\ & = \frac{7}{8} \end{aligned}$$

---

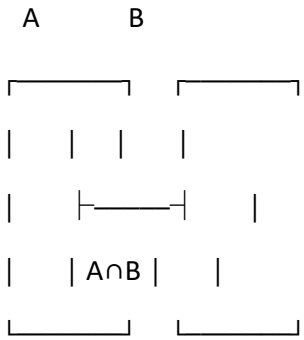
5. Addition Rule

For arbitrary events:

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

The intersection must be subtracted because it gets counted twice.

Diagram



Example

Suppose:

P(A) = 0.6

P(B) = 0.5

P(A∩B) = 0.2

Then:

$$P(A \cup B) = 0.6 + 0.5 - 0.2 = 0.9$$

Important distinction

If A and B are **mutually exclusive**:

$$P(A \cap B) = 0$$

Therefore:

$$P(A \cup B) = P(A) + P(B)$$

---

6. Conditional Probability ⭐ ⭐ ⭐

This is **extremely important for ML interviews**.

Definition

Probability of B given that A has occurred:

$$\boxed{P(B \mid A) = \frac{P(A \cap B)}{P(A)}}$$

provided:

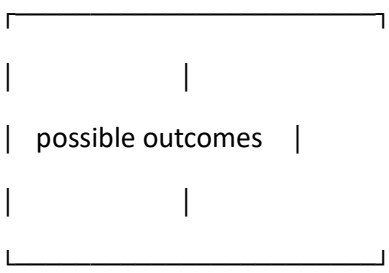
$$P(A) > 0$$

The source gives this as its formal definition of conditional probability.

Intuition

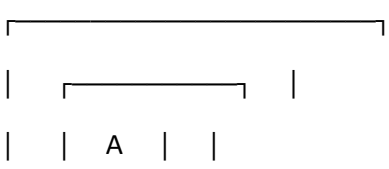
Without information:

Ω



After learning A:

Ω



| | possible | |

| | outcomes | |

| └──────────┘ |

└──────────────────────┘

We **restrict our attention to A**.

Example

A die is rolled.

Given that the result is odd:

Possible outcomes = {1,3,5}

Probability of getting 3:

$$P(3 \mid odd) = \frac{1}{3}$$

The source works through exactly this dice example.

---

7. Multiplication Rule

From conditional probability:

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

Therefore:

$$\boxed{P(A \cap B) = P(B \mid A)P(A)}$$

Also:

$$P(A \cap B) = P(A \mid B)P(B)$$

Why important?

This is the foundation for:

- Bayesian inference
- Naive Bayes

- probabilistic graphical models
- likelihood calculations
- sequential probability

The source explicitly develops this into the general chain rule.

---

8. Independence ⭐⭐⭐

Definition

Two events are independent if occurrence of one doesn't change the probability of the other.

Mathematically:

$$\boxed{P(A \cap B) = P(A)P(B)}$$

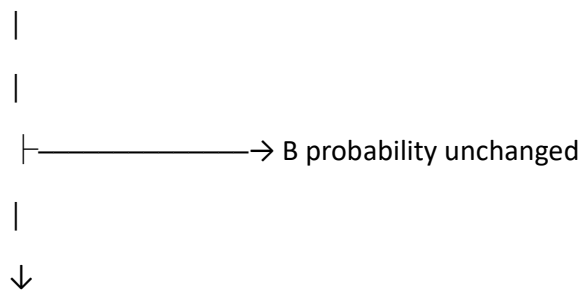
The source gives this exact definition.

Equivalent form:

$$P(B \mid A) = P(B)$$

Diagram

A happens



No information about B

Example

Two fair coin tosses.

P(H on toss 1) = 1/2

P(H on toss 2) = 1/2

Therefore:

$$P(H_1 \cap H_2) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

The source uses this example and emphasizes that even after ten heads, the probability of heads on the next independent toss remains 50%.

Interview trap

Independent ≠ mutually exclusive

If two events are mutually exclusive and both have nonzero probability:

$$P(A \cap B) = 0$$

But independence would require:

$$P(A \cap B) = P(A)P(B) > 0$$

So they cannot be both independent and mutually exclusive unless one has probability zero.

---

9. Law of Total Probability ⭐ ⭐ ⭐

This is important because it lets you break a complicated probability into simpler cases.

Suppose:

Ω

├— A_1

├— A_2

├— A_3

└— ...

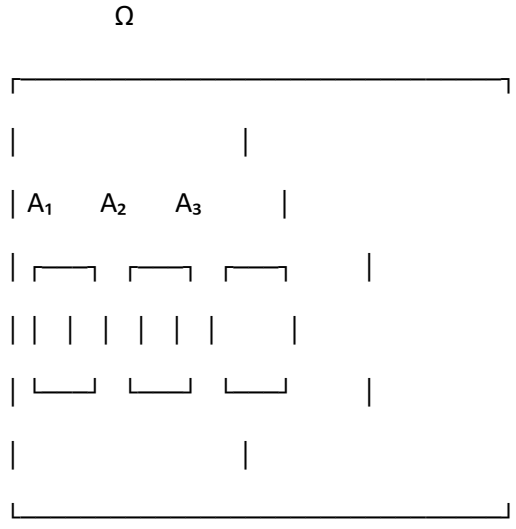
where the events form a partition.

Then:

$$\boxed{P(B) = \sum_i P\left(B \mid A_i\right) P(A_i)}$$

The source describes the partition as mutually disjoint events whose union is the entire event space.

Diagram



P(B)

$= P(B|A_1)P(A_1)$

$+ P(B|A_2)P(A_2)$

$+ P(B|A_3)P(A_3)$

ML example

Suppose emails come from:

Normal users: 80%

Spammers: 20%

Probability an email contains suspicious word:

P(word | normal) = 0.01

P(word | spam) = 0.60

Then:

$$P(word) = 0.01(0.8) + 0.60(0.2)$$
$$= 0.008 + 0.12$$
$$= 0.128$$

So approximately **12.8%** of emails contain the word.

---

10. Bayes' Theorem ⭐ ⭐ ⭐ ⭐ ⭐

For ML interviews, this is one of the concepts you should be able to explain without hesitation.

Formula

$$\boxed{P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A)}}$$

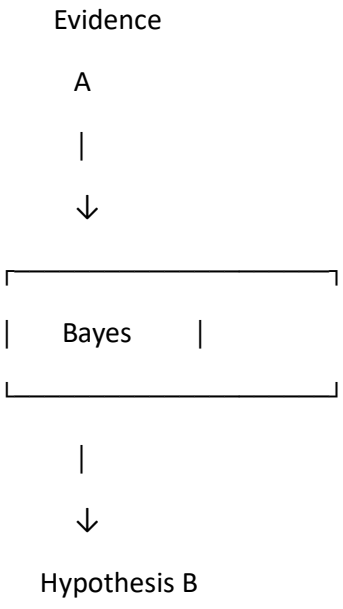
Using total probability:

$$\boxed{P(B \mid A) = \frac{P(A \mid B)P(B)}{\sum_i P\,(A \mid B_i)P(B_i)}}$$

The uploaded material presents Bayes' theorem as a way to reverse the direction of conditional probability and later interprets it as updating a probabilistic model using new observations.

Bayes Diagram

The most useful way to remember it:



P(B|A)

=

P(A|B) × P(B)

_____

P(A)

Remember:

Posterior

=

Likelihood × Prior

_____

Evidence

---

11. Bayes Example — Medical Test

Suppose:

P(Disease) = 0.02

P(Positive | Disease) = 0.99

P(Positive | Healthy) = 0.05

We want:

$$P(Disease \mid Positive)$$

First calculate:

$$P(Positive) = P(Positive \mid Disease)P(Disease) + P(Positive \mid Healthy)P(Healthy)$$
$$= 0.99(0.02) + 0.05(0.98)$$
$$= 0.0198 + 0.049$$
$$= 0.0688$$

Then:

$$P(Disease \mid Positive) = \frac{0.99(0.02)}{0.0688}$$
$$\approx 0.288$$

So approximately:

$$\boxed{28.9\%}$$

The uploaded source uses this same example and obtains approximately 29%.

Interview lesson

A positive test does **not automatically mean** the disease probability is 99%.

You must consider the **base rate/prior**.

---

12. Prior, Likelihood, Posterior

These three terms are extremely important.

Prior

Belief before seeing new evidence.

$$P(Disease)$$

Likelihood

How probable the observed evidence is under the hypothesis.

$$P(Positive \mid Disease)$$

Posterior

Updated belief after seeing evidence.

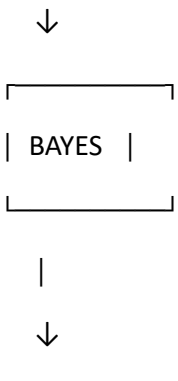
$$P(Disease \mid Positive)$$

Mental model

PRIOR

|

| + evidence



The source explicitly describes the posterior as the result of combining likelihood with the prior.

---

13. Chain Rule of Probability

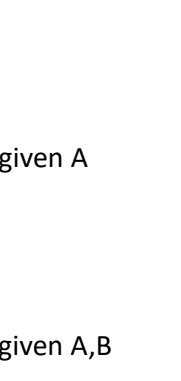
For multiple events:

$$P(A_1, A_2, \dots, A_n) = \prod_{k=1}^{n} P\,(A_k, \mid, A_1 \dots A_{k-1})$$

For three events:

$$P(A, B, C) = P(A)P(B \mid A)P(C, \mid AB)$$

Diagram



Example

Probability of:

First toss = H

Second toss = H

Third toss = T

$$P(H,H,T) = P(H)P(H \mid H)P(T, \mid HH)$$

For independent fair tosses:

$$= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$$

---

14. Random Variable

This is the bridge from probability to statistics and ML.

A **random variable assigns a numerical value to an outcome of a random experiment.**

Example:

Experiment: roll a die

Outcome → Random variable X

1 → X=1

2 → X=2

3 → X=3

...

6 → X=6

Diagram

Sample Space Ω

{1,2,3,4,5,6}

|

| X

↓

Numerical value

In ML, observations are often represented as vectors:

$$X \in \mathbb{R}^n$$

The source later connects probability spaces to ML data points in Euclidean space.

---

15. Discrete vs Continuous Random Variables

Discrete

Countable values.

Examples:

Number of customers

Number of clicks

Number of defects

Dice result

Continuous

Can take values over an interval.

Examples:

Height

Weight

Temperature

Response time

Important distinction

For continuous variables:

$$P(X = x) = 0$$

can still be true even though X=x is possible.

The uploaded source demonstrates this using a uniformly selected number from [0,1].

Instead, we calculate probabilities over ranges:

$$P(a < X < b)$$

---

16. Probability Distribution

A probability distribution describes how probability is assigned across possible values of a random variable.

Example:

X = dice roll

X 1 2 3 4 5 6

P(X) 1/6 1/6 1/6 1/6 1/6 1/6

Diagram

Probability

|

1/6| ▇▇▇▇▇▇

|▇▇▇▇▇▇

└──────────

1 2 3 4 5 6

X

---

17. Expected Value ⭐⭐⭐

The expected value is the probability-weighted average.

For discrete variables:

$$\boxed{E[X] = \sum_x x\, P(X = x)}$$

Dice example

$$E[X] = 1\frac{1}{6} + 2\frac{1}{6} + 3\frac{1}{6} + 4\frac{1}{6} + 5\frac{1}{6} + 6\frac{1}{6}$$
$$= \frac{21}{6} = 3.5$$

Critical interview point

Expected value does NOT mean the result you should expect to actually observe.

You cannot roll a die and get 3.5.

It means:

The long-run average outcome approaches 3.5 under repeated independent rolls.

---

18. Variance ⭐ ⭐ ⭐

Variance measures how spread out values are around the mean.

$$\boxed{Var(X) = E[(X - E[X])^2]}$$

Equivalent:

$$\boxed{Var(X) = E[X^2] - (E[X])^2}$$

Intuition

Low variance:



mean

High variance:



mean

ML importance

Variance is fundamental to:

- model evaluation

- noise
- bias-variance tradeoff
- distributions
- feature analysis
- uncertainty
- statistical estimation

---

19. Standard Deviation

$$\boxed{\sigma = \sqrt{Var(X)}}$$

Why square root?

Because variance is measured in squared units.

Example:

Height → meters

Variance → meters²

Standard deviation → meters

So standard deviation is easier to interpret in the original units.

---

20. Law of Large Numbers

This is the intuition behind why empirical frequencies approach probability.

Suppose a fair coin is repeatedly tossed.

Tosses Heads proportion

10 ~0.6

100 ~0.53

1,000 ~0.50

10,000 ~0.50

As the number of observations increases:

$$\text{empirical frequency} \rightarrow P(A)$$

The source describes this frequentist interpretation and shows the relative frequency stabilizing around 0.5 for repeated fair coin tosses.

---

21. Why Probability Matters in Machine Learning

This is the **interview connection you should memorize conceptually**:

Real world

|

↓

Uncertainty

|

↓

Probability

|

↓

Statistical model

|

↓

Learn from data

|

↓

Prediction

For classification:

$$P(Y \mid X)$$

means:

Probability of label Y given observed features X.

For example:

X =

age = 35

income = ₹10L

previous_purchase = yes

↓

Model

↓

P(buy | X) = 0.82

This probabilistic formulation is explicitly highlighted in the source as a central ML idea.

---

22. Implementation — Python

Because you're preparing for AI/ML engineering interviews, you should be able to implement the fundamental rules yourself before relying on libraries.

Basic probability

```
def probability(favorable, total):
    if total == 0:
        raise ValueError("Total cannot be zero")
    return favorable / total



print(probability(3, 6))
# 0.5
```

Complement

```
def complement_probability(p):
    if not 0 <= p <= 1:
        raise ValueError("Probability must be between 0 and 1")
    return 1 - p



print(complement_probability(0.2))
# 0.8
```

Addition rule

```
def union_probability(p_a, p_b, p_a_and_b):
    return p_a + p_b - p_a_and_b



print(union_probability(0.6, 0.5, 0.2))
# 0.9
```

Conditional probability

```
def conditional_probability(p_a_and_b, p_a):
    if p_a == 0:
        raise ValueError("P(A) must be greater than zero")

    return p_a_and_b / p_a



print(conditional_probability(0.2, 0.5))
# 0.4
```

Independence check

```
def are_independent(p_a, p_b, p_a_and_b):
    return abs(p_a_and_b - p_a * p_b) < 1e-12



print(are_independent(
    p_a=0.5,
    p_b=0.5,
    p_a_and_b=0.25
))
# True
```

Bayes theorem

```
def bayes(p_a_given_b, p_b, p_a):
    if p_a == 0:
        raise ValueError("P(A) must be greater than zero")

    return (p_a_given_b * p_b) / p_a



# Medical example
posterior = bayes(
    p_a_given_b=0.99,   # P(positive | disease)
    p_b=0.02,           # P(disease)
    p_a=0.0688          # P(positive)
)

print(posterior)
```

```
# approximately 0.288
```

Better implementation

In an interview, calculate the denominator explicitly:

```
p_disease = 0.02
p_healthy = 1 - p_disease

p_positive_given_disease = 0.99
p_positive_given_healthy = 0.05

p_positive = (
    p_positive_given_disease * p_disease
    + p_positive_given_healthy * p_healthy
)

p_disease_given_positive = (
    p_positive_given_disease * p_disease
    / p_positive
)

print(p_disease_given_positive)
# 0.28779...
```

This implementation directly reflects the Bayes calculation in your source.

---

23. A Small Probability Engine

For interview practice, you can combine the concepts:

```
class Probability:
    @staticmethod
    def complement(p):
```

```
        return 1 - p

    @staticmethod
    def union(p_a, p_b, p_intersection):
        return p_a + p_b - p_intersection

    @staticmethod
    def conditional(p_intersection, p_given):
        if p_given <= 0:
            raise ValueError("Conditioning probability must be > 0")
        return p_intersection / p_given

    @staticmethod
    def joint(p_a, p_b_given_a):
        return p_a * p_b_given_a

    @staticmethod
    def bayes(p_a_given_b, p_b, p_a):
        if p_a <= 0:
            raise ValueError("Evidence probability must be > 0")
        return p_a_given_b * p_b / p_a


print(Probability.complement(0.2))
print(Probability.union(0.6, 0.5, 0.2))
print(Probability.conditional(0.2, 0.5))
print(Probability.joint(0.5, 0.5))
print(Probability.bayes(0.99, 0.02, 0.0688))
```

The important thing isn't writing this class in production. The value is being able to **derive every operation from the definitions**.

---

24. Interview Cheat Sheet

Concept	Formula	Interview importance
Probability	$0 \le P(A) \le 1$	⭐⭐⭐
Complement	$P(A^c)=1-P(A)$	⭐⭐⭐
Addition	$P(A \cup B)=P(A)+P(B)-P(A \cap B)$	⭐⭐⭐⭐
Conditional probability	$P(B \mid A)=P(A \cap B)/P(A)$	⭐⭐⭐⭐⭐
Joint probability	$P(A \cap B)=P(B \mid A)P(A)$	⭐⭐⭐⭐⭐
Independence	$P(A \cap B)=P(A)P(B)$	⭐⭐⭐⭐⭐
Total probability	$\Sigma P(B \mid Ai)P(Ai)$	⭐⭐⭐⭐⭐
Bayes	$P(B \mid A)=P(A \mid B)P(B)/P(A)$	⭐⭐⭐⭐⭐
Expected value	$\Sigma x P(x)$	⭐⭐⭐⭐⭐
Variance	$E[X^2]-E[X]^2$	⭐⭐⭐⭐⭐
Std deviation	$\sqrt{}$variance	⭐⭐⭐⭐
LLN	empirical frequency → probability	⭐⭐⭐⭐

---

25. The 5 Formulas You Should Know Cold

If your interview is close, prioritize these:

1. Conditional probability

$$\boxed{P(B \mid A) = \frac{P(A \cap B)}{P(A)}}$$

2. Multiplication

$$\boxed{P(A \cap B) = P(B \mid A)P(A)}$$

3. Independence

$$\boxed{P(A \cap B) = P(A)P(B)}$$

4. Total probability

$$\boxed{P(B) = \sum_i P\,(B \mid A_i)P(A_i)}$$

5. Bayes

$$\boxed{P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A)}}$$

And remember the conceptual chain:

Conditional probability

↓

Multiplication rule

↓

Law of total probability

↓

Bayes theorem

↓

Bayesian inference / ML

---

26. Common Interview Traps

Trap 1 — Confusing P(A|B) and P(B|A)

They are generally **not equal**.

P(disease | positive)

≠

P(positive | disease)

Bayes theorem exists partly because we need to move between these quantities.

Trap 2 — Ignoring the base rate

A rare disease can have a highly accurate test and still produce many false positives relative to true positives.

This is exactly what the source's medical example demonstrates.

Trap 3 — Thinking independent means mutually exclusive

Wrong:

Independent = cannot happen together

Correct:

Independent = occurrence of A does not change probability of B

Trap 4 — Gambler's fallacy

If a fair coin gives:

H H H H H H H H H H

the probability of the next H is still:

$$P(H) = 0.5$$

assuming independent fair tosses.

Trap 5 — Using favorable/total blindly

$$P(A) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$$

works directly when outcomes are **equally likely**. The source explicitly gives this caveat.

---

27. What to Prioritize for Your Interview

Using the **80/20 principle**, don't spend equal time on every mathematical detail in this chapter.

P0 — Must master

Probability basics

↓

Events / sample space

↓

Complement

↓

Union / intersection

↓

Conditional probability

↓

Independence

↓

Multiplication rule

↓

Total probability

↓

Bayes theorem

↓

Expected value

↓

Variance / standard deviation

P1 — Understand well

Random variables

Probability distributions

Discrete vs continuous

Law of Large Numbers

Joint probability

Chain rule

Bayesian interpretation

P2 — Don't over-invest initially

σ-algebra construction

Generated σ-algebras

Borel algebra proofs

Measure-theoretic technicalities

Continuity proofs

These are mathematically important, but for an **ML engineering interview**, the source's deeper measure-theoretic material is usually lower-return than mastering conditional probability, Bayes, independence, distributions, expectation, and variance.

The uploaded source spends substantial space formally defining σ-algebras and Borel sets, but the practical ML connection comes later when probability is used to model data and uncertainty.

---

28. Final Mental Model

If an interviewer asks:

"Why do we need probability in machine learning?"

Build your answer from this:

Real-world data

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Incomplete information

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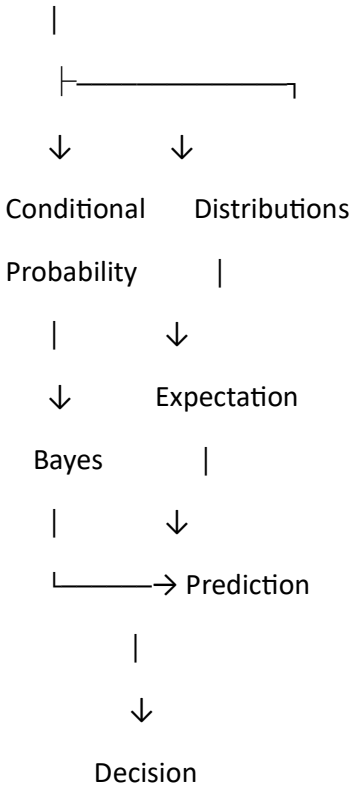
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Uncertainty

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Probability



The core idea from your source is that ML isn't merely about outputting a label; it is about constructing a mathematical model that represents knowledge about data **and the degree of certainty in predictions**.

Your interview-ready one-minute answer

Probability is a mathematical framework for representing uncertainty. An experiment has a sample space of possible outcomes, and events are subsets of that space to which we assign probabilities between 0 and 1. Conditional probability lets us update the probability of an event given evidence, independence tells us when one event does not affect another, the law of total probability decomposes complex probabilities into cases, and Bayes' theorem reverses conditional probabilities using prior knowledge and evidence. In machine learning, this becomes especially important because prediction can be expressed as estimating probabilities such as P(label | data), allowing models to represent uncertainty rather than only output a hard class label.

That is the **80/20 core** I would make sure you can explain, derive, calculate, and implement before moving deeper into the remaining probability material.